

Course Title: Introduction to Mechanical Engineering

Course Code: BESCK104D/204D

Course In charge: Ashoka

Module-5

Introduction to Mechatronics and Robotics: Open-loop and closed-loop mechatronic systems.

Classification based on robotics configuration: polar cylindrical, Cartesian coordinate and spherical.

Application, Advantages and disadvantages. **Automation in industry:** Definition, types – Fixed, programmable and flexible automation, basic elements with block diagrams, advantages. **Introduction to**

IOT: Definition and Characteristics, Physical design, protocols, Logical design of IOT, Functional blocks, and communication models.

Control system

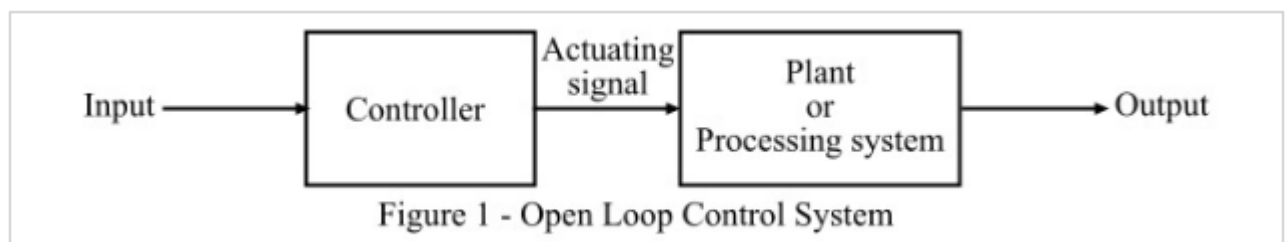
A control system is defined as the mechanism or a system that provides the desired response or output by controlling the input and processing system. These days, control systems are being implementing in every smart devices and systems such as traffic lights system, washing machine, rocket launching systems, electrical power system, etc.

Depending on the feedback path present in the system, the control systems can be classified into following two types

- Open Loop Control System
- Closed Loop Control System

Open Loop Control System

An **open loop control system** is the one in which the output signal is not fed back to the input of the system. Therefore, an open loop control system is also referred to as a non-feedback control system.



In case of open loop control system, the output has no control on the control action of the system. Thus, the open loop control system follows its input signals regardless of the final results. The input is supplied to the controller which produces an actuating signal (or control signal). This actuating single is supplied to the plant or processing system which is to be controlled.

Practical Examples of Open Loop Control System



Fig.2 Hand drier [III]

Electric Hand Drier

Washing Machine

Bread Toaster

Light Switch

Volume on Stereo System



Fig.3 Washing machine [III]



Fig.4 Toaster [IV]



Advantages of Open Loop Control System

- Simple in construction and design
- Economical
- Easy to maintain
- Generally stable
- Convenient to use as output is difficult to measure

Disadvantages of Open Loop Control System

- They are inaccurate
- They are unreliable
- Any change in output cannot be corrected automatically

Closed Loop Control System

A closed loop control system is the one in which the output signal is fed-back to the input of the system. Therefore, in a closed loop control system, the control action is a function of desired output signal.

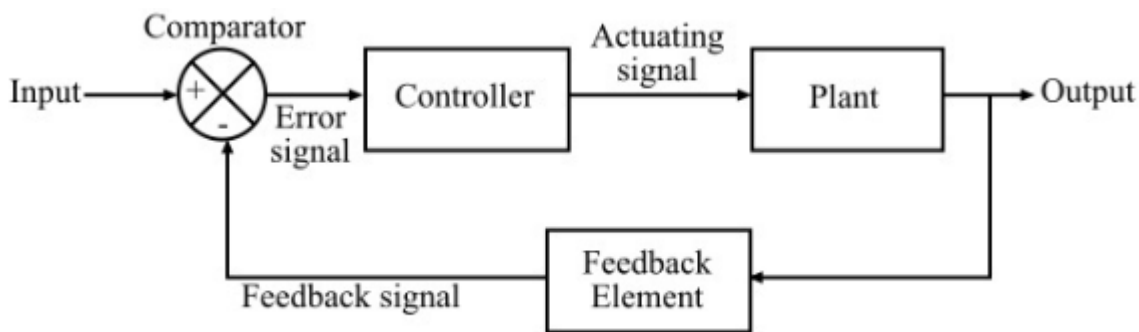


Figure 2 - Closed Loop Control System

The error detector accepts input signal and feedback signal to produce an error signal which is the difference of input and feedback signals. The feedback signal is the sample of output of the overall system.

Now, the error signal is supplied to the controller to produce an actuating signal which controls the plant or processing system to produce desired results. Therefore, in the closed loop control system, the input of the system is automatically adjusted to produce a desired response from the system.

Practical Examples of Closed Loop Control System



Fig.8 Water level controller [VII]



Fig.9 Electric Iron [VIII]

Water Level Controller

Automatic Electric Iron

Voltage Stabilizer

Air Conditioner



Fig.10 Voltage stabilizer [X]



Fig.11 Air conditioner [XI]

Advantages of Closed Loop Control System

- more accurate even in the presence of non-linearity.
- Highly accurate as any error arising is corrected due to presence of feedback signal.
- Bandwidth range is large.
- Facilitates automation.
- The sensitivity of system may be made small to make system more stable.
- less affected by noise.

Disadvantages of Closed Loop Control System

- They are costlier.
- They are complicated to design.
- Required more maintenance.
- Feedback leads to oscillatory response.
- Overall gain is reduced due to presence of feedback.
- Stability is the major problem and more care is needed to design a stable closed loop system.

Difference between Open-Loop Control System and Closed-Loop Control System

Basis of Difference	Open Loop Control System	Closed Loop Control System
Definition	A control system in which there is no feedback path is provided is called an open loop control system.	The control system in which there is a feedback path present is called a closed loop control system.
feedback control system	Open loop control system is also called non-feedback control system	Closed loop control system is also called a feedback control system
Control action	In open loop control system, the control action is independent of the output of the overall system.	In closed loop control system, the control action is dependent on the output of the system.
Design complexity	The design and construction of an open loop control system is quite simple.	Closed loop control system has comparatively complex design and construction.
Main Components	The major components of an open loop control system are – controller and plant.	The main components of a closed loop control system are – Controller, plant or process, feedback element and error detector (comparator).
Response	Open loop control system has fast response because there is no measurement and feedback of output.	The response of the closed loop control system is slow due to presence of feedback.
Reliability	The reliability of open loop control system is less.	The closed loop control system is more reliable.
Accuracy	The accuracy of open loop control system depends upon the system calibration and therefore, may be less.	Closed loop control system is comparatively accurate because the feedback maintains its accuracy.
Maintenance	Open loop control system requires less maintenance.	Comparatively more maintenance is needed in closed loop control system.
Implementation	Open loop control system is easy to implement.	The implementation of a closed loop control system is relatively difficult.

Cost	Open loop control system is less expensive.	The cost of the closed loop control system is relatively high.
Noise	Open loop control system has more internal noise.	In closed loop system, the internal noise in the system is less.
Examples	Electric Hand Drier, Washing Machine, Bread Toaster, Light switch etc.	Air Conditioner, Water Level controller, Automatic Electric Iron, Voltage stabilizer, etc.

INDUSTRIAL ROBOTICS

Robot Anatomy (Optional)

ROBOT ANATOMY AND RELATED ATTRIBUTES

- The anatomy of industrial robots deals with the assembling of outer components of a robot such as wrist, arm and body.
- Before jumping into robot configurations, here are some of the key facts about robot anatomy.

(a) Joints and Links

(b) Common Robot Configurations

JOINTS AND LINKS

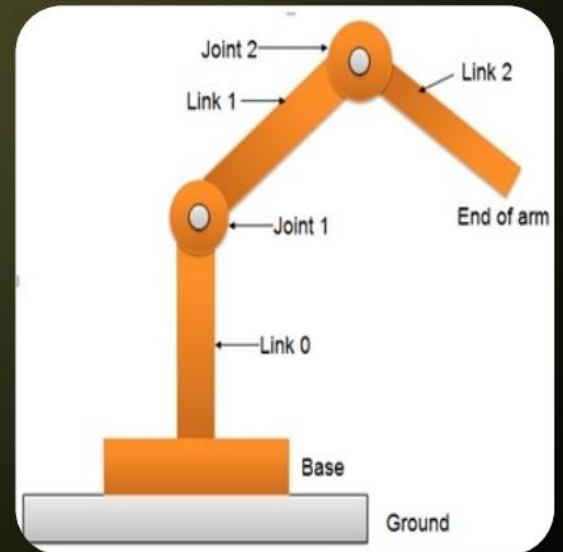
- The manipulator of an industrial robot consists of a series of joints and links.
- Robot anatomy deals with the study of different joints and links and other aspects of the manipulator's physical construction.
- A robotic joint provides relative motion between two links of the robot.
- Each joint, or axis, provides a certain degree-of-freedom (dof) of motion.
- In most of the cases, only one degree-of-freedom is associated with each joint.
- Robot's complexity can be classified according to the total number of degrees-of-freedom they possess.
- Each joint is connected to two links, an input link and an output link.

JOINTS AND LINKS

- A Joint provides controlled relative movement between the input link and output link. A robotic link is the rigid component of the robot manipulator.
- Most of the robots are mounted upon a stationary base, such as the floor. From this base, a joint-link numbering scheme may be recognized as shown in Figure.

JOINTS AND LINKS

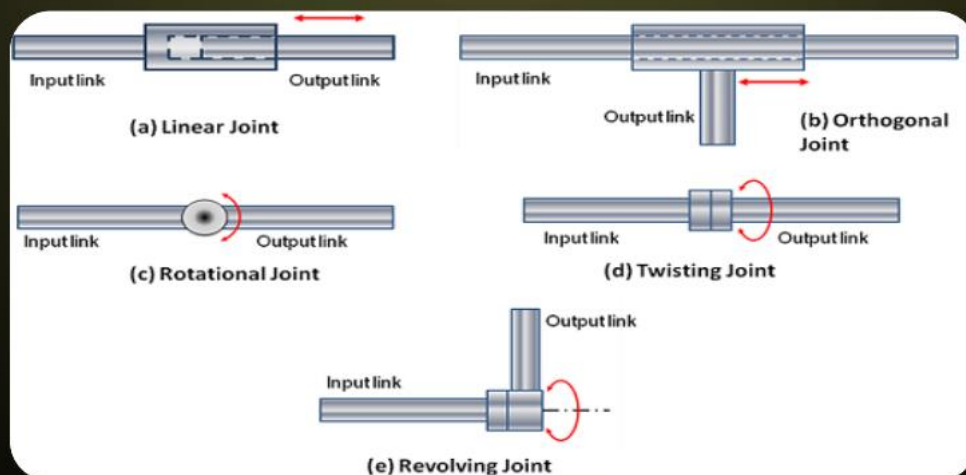
- The robotic base and its connection to the first joint are termed as link-0.
- The first joint in the sequence is joint-1.
- Link-0 is the input link for joint-1, while the output link from joint-1 is link-1 which leads to joint-2.
- Link 1 is the output link for joint-1 and the input link for joint-2.
- This joint-link-numbering scheme is further followed for all joints and links in the robotic systems.



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JOINTS AND LINKS

- Nearly all industrial robots have mechanical joints that can be classified into following five types as shown in Figure below.



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JOINTS AND LINKS

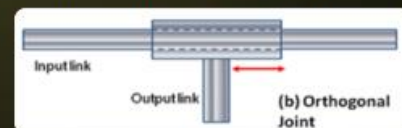
a) Linear joint (type L joint)

- The relative movement between the input link and the output link is a translational sliding motion, with the axes of the two links being parallel.



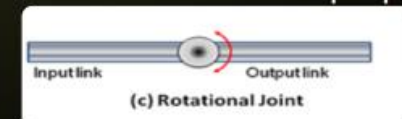
b) Orthogonal joint (type U joint)

- This also has a translational sliding motion, but the input and output links are perpendicular to each other during the move.



c) Rotational joint (type R joint)

- This type provides rotational relative motion, with the axis of rotation perpendicular to the axes of the input and output links.

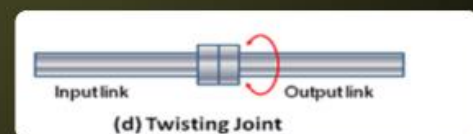


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JOINTS AND LINKS

d) Twisting joint (type T joint)

- This joint also involves rotary motion, but the axis of rotation is parallel to the axes of the two links.



e) Revolving joint (type V-joint, V from the "v" in revolving)

- In this type, axis of input link is parallel to the axis of rotation of the joint. Axis of the output link is perpendicular to the axis of rotation.



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Robot Configuration

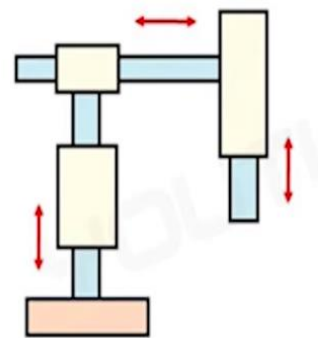
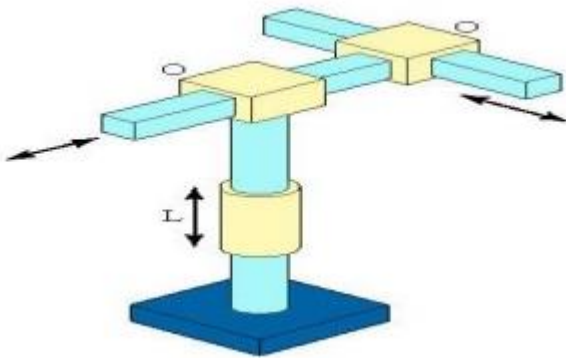
Industrial robots are available in a wide variety of sizes, shapes, and physical configurations. These configurations are mainly depending on the movement of joints.

Classification based on robotics configuration:

1. Cartesian Robot Configuration (3P)
2. Polar Configuration (P2R) or Spherical configuration
3. Cylindrical robot configuration
4. Jointed-arm configuration

1. Cartesian Robot Configuration (3P)

These robots are made of three linear joints that position the end effector, which is usually followed by additional revolute joints that orientate the end effector. 3P represents three Prismatic or Linear joints used in the robots. By moving the three slides relative to one another, the robot is capable of operating within a rectangular work envelope.



Advantages

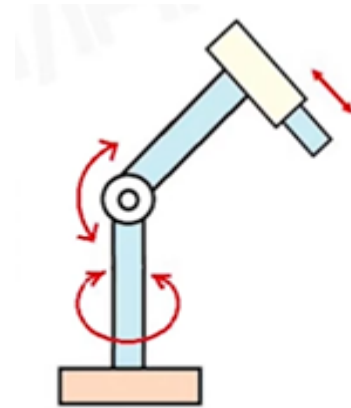
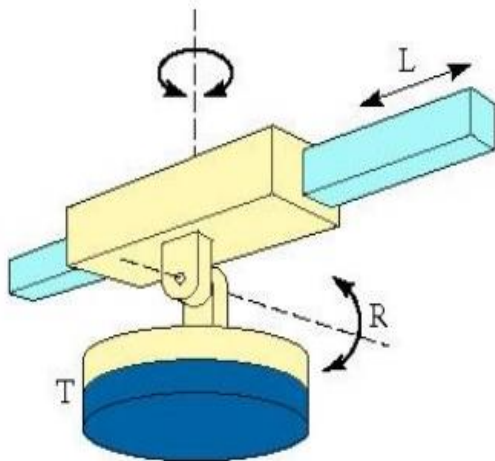
- Highly accurate & speed,
- Fewer costs,
- Simple operating procedures, and
- High payloads.

Disadvantages

- Less work envelope, reduced flexibility.
- Low dexterity (not able to move quickly and easily)
- Limited manipulability

2. (i) Polar Configuration (P2R)

The polar configuration uses a telescopic arm that can be moved up or down about a horizontal pivot. The pivot is mounted on a rotating base. A rotating base supports the two links. These various joints can move the arm endpoint within a partial spherical space. It is also called a spherical co-ordinate configuration, which allows manipulation of objects on the floor. This system has one Prismatic and two revolute joints that denote the P2R.



Advantages

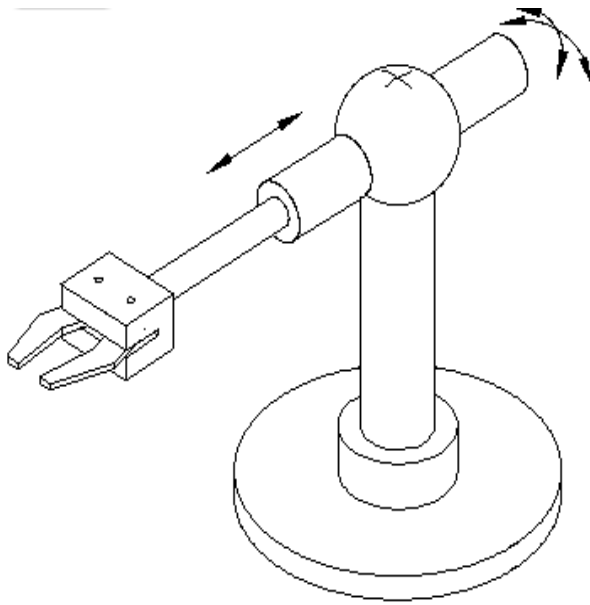
- Long reach capability in the horizontal position.

Disadvantages

- Vertical reach is low
- Low mechanical stiffness
- Complex construction
- Accuracy decreases with the increasing radial stroke.

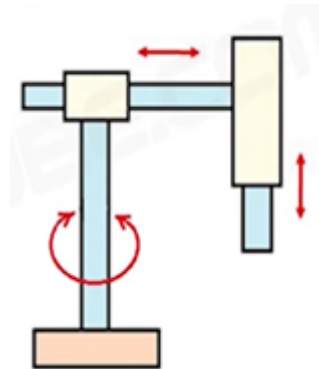
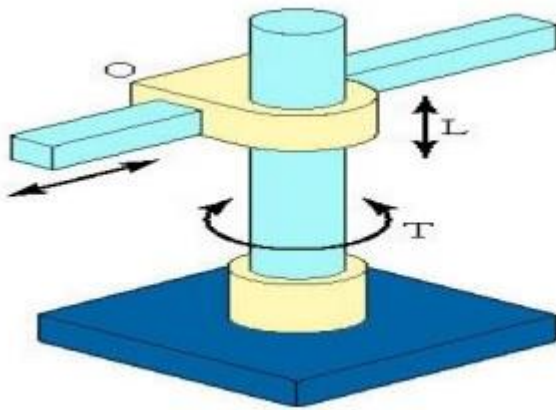
ii) Spherical configuration

A spherical robot is a robot with two rotary joints and one prismatic joint; in other words, two rotary axes and one linear axis. Spherical robots have an arm which forms a spherical coordinate system.



3. Cylindrical Robot Configuration (PRP)

The cylindrical configuration uses a vertical column and a slide that can be moved up or down along the column. The robot arm is attached to the slide so that it can be moved radially with respect to the column. By rotating the column, the robot is capable of achieving a workspace that approximates a cylinder. This cylindrical configuration has two Prismatic or Linear joints and one revolute joint, (PRP) denotes this only.



Advantages

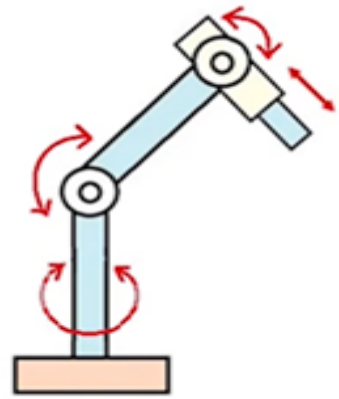
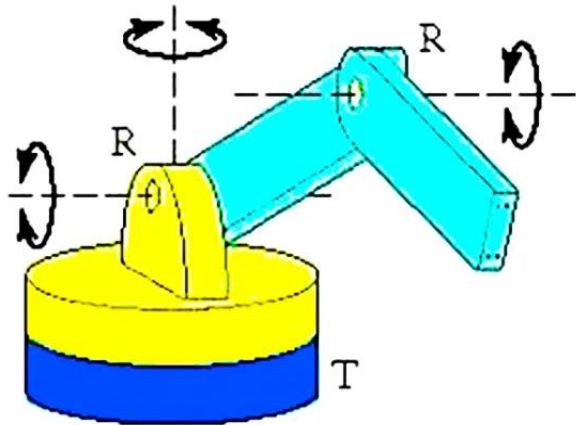
- Increased rigidity, and
- Capacity to carry high payloads

Disadvantages

- Floor space required is more, and
- Less work volume.
- Accuracy decreases as the horizontal stroke increases.

4. Jointed Arm Configuration

The jointed Arm configuration is made up of rotating joints. This robot configuration is also sometimes called anthropomorphic as its anatomy is similar to the Human arm. These configuration joints are all revolute (3R). They are the most common configuration for industrial robots.



Advantages

- Increased flexibility,
- Huge work volume, and
- Quick operation.

Disadvantages

- Very expensive,
- Difficult operating procedures, and
- Plenty of components.

Advantages and Disadvantages of Robots

Advantages of Robots

- Increase in productivity, safety, efficiency, quality and consistency of product.
- Can work in hazardous environments
- Need no environmental comfort.
- Repeatable precision at all times.
- More accurate than humans; they may have milli or micro inch accuracy.
- Capabilities beyond that of humans.
- Process multiple tasks simultaneously

Disadvantages of Robots

- Robots lack capability to respond in emergencies.
- A lack of decision making power
- Degrees of freedom
- Real – time response
- Initial cost of equipment
- Installation cost
- Need for training

Applications of Robots

Applications

(i) Material Handling

- Used for pick & place operation.
- To stack parts one on top of the other, that is to palletize them
- Parts are unloaded by robots.
- Unloading the parts from the machine assisted by vision or no vision.
- Robot loads a raw work part into the process and unloads a finished part.

(ii) Processing

- ✓ Spot welding by squeezing the metal parts together using electrodes.
- ✓ Continuous arc welding for joining the metals using heat generated by electric arc.
- ✓ Spray painting is done by spraying the fluid on metal parts.
- ✓ Metal cutting and deburring operations
- ✓ Various machining operations like drilling, grinding, laser and water-jet cutting and riveting are also carried.
- ✓ Rotating and spindle operations
- ✓ Adhesives and sealant dispensing

(iii) Assembly & Inspection

- ✓ To detect flaws by comparing the good part with bad part.
- ✓ Need to replace human labor by robot.
- ✓ Work environment hazardous for human beings.
- ✓ Repetitive task
- ✓ Multi shift operations
- ✓ Operating for long hours without rest

Automation

Automation could be defined as the technology dealing with design of machine tools and systems utilising electronics and computer systems to produce the final product at minimum cost, involving minimum labour intervention, producing components of high accuracy repeatedly without causing rejections.

The Advantages of automation are:

- ❖ Reduced human labour required
- ❖ Consistency in quality
- ❖ Fewer risks of human error
- ❖ Improved health & safety
- ❖ Improved efficiency

The Disadvantages of automation are:

- ❖ Initial investment is high
- ❖ Require some manual labour: observation and programming
- ❖ Failures could lead to temporary downtime in production

Types of Manufacturing Automation

- 1) Fixed Automation
- 2) Programmable Automation
- 3) Flexible Automation

1) Fixed Automation

In fixed automation, the sequence of processing is fixed for one particular product. Such a system can be used for mass production of a product. Its initial cost is high and any changes in product design can be incorporated with difficulty.

If product has to be changed, then lot of modifications, new additions, etc. may be required and to accommodate same would be very costly and time consuming. But it provides high production rates. This type of automation is best suited for: High demand and general products that require no change.

This is relatively useful for many companies who use automation to create food products of one type and variant. It allows them to effectively produce that item and package it in bulk.

Advantages of fixed automation include:

- ❖ High levels of production
- ❖ Consistent quality in production
- ❖ Low cost per unit produced

Disadvantages of fixed automation include:

- ❖ High initial cost
- ❖ Difficult to accommodate change

2. Programmable Automation

In programmable automation (as with numerically controlled machines and robots), it is possible to accommodate the change in sequence of operations for new product by changing the program (set of instructions). Such a system is thus suited for batch production. For new product, not only programs have to be changed, but new tools and fixtures may have to be loaded, and machine settings changed.

Programmable automation will often be used by factories who make different variants of foods. This allows them to make batches, from a few dozen to potentially thousands at a time, of one product. If the product needs changing, it simply needs to be reprogrammed.

This type of automation is well suited for: Low/Medium demand and occasional changes in products.

Advantages of Programmable automation:

- ❖ Flexibility to change products if needed
- ❖ Suitable if batch production is required

Disadvantages of Programmable automation:

- ❖ Expensive for equipment
- ❖ Lower production levels
- ❖ Often time-consuming to change products

3. Flexible Automation

Flexible automation, also known as “soft automation”, is similar to programmable automation, although a little more complicated. Essentially, flexible automation enables the production of different types of products without losing time when reprogramming.

A flexible automation system can produce various combinations of products efficiently without having to separate them into different batches, as required in batch production. No time is lost for production of one product and changing over to new product. Complete information and programs for the products desired to be produced are available in computer system

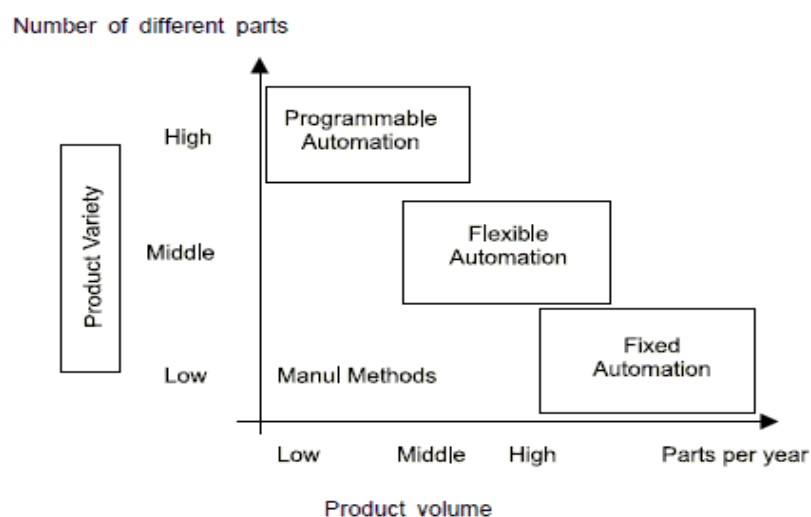
Altering the part programme is usually done by coding the program offline on a computer and then transferring it to the automated system. Such systems are best suited for production of large variety of spare parts for large industrial plant/process/complex.

Advantages of automation:

- ❖ Flexibility of products
- ❖ No time lost with new changes to production

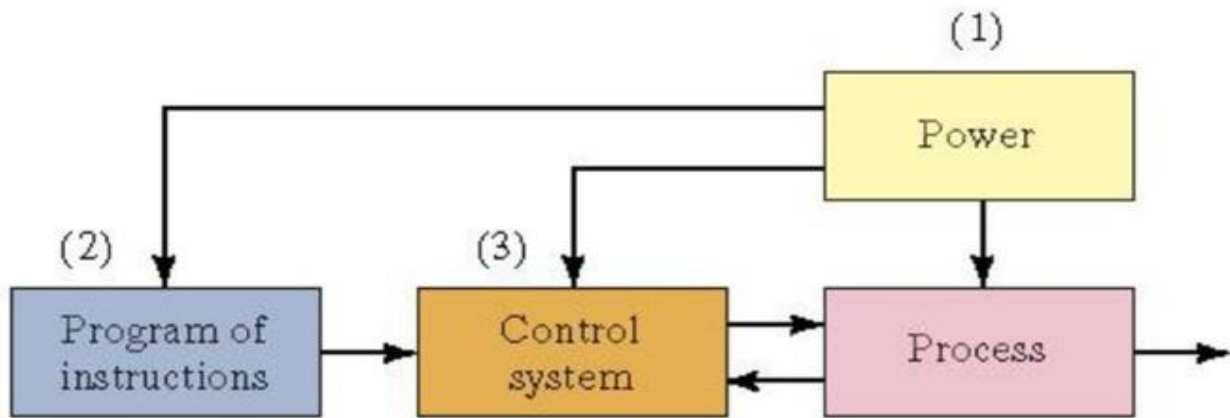
Disadvantages include:

- ❖ High custom machinery/automation cost
- ❖ Higher cost per unit



types-of-production-automation

Basic Element of an Automated System



1.5 Basic elements of an automated system -

Basic elements of an automated system:

1. Power to accomplish the process and operate the system.
2. A program of instructions to direct the process.
3. A control system to actuate the instructions.

Elements of Automated system

1. Power source: An automated system is used to operate some process and power is required to drive the process as well as controls. There are many sources of power available, but the most commonly used power is electricity. The actions performed by automated systems are generally of two types:

(a) Processing

(b) Transfer and positioning

In the first case, energy is applied to accomplish some processing operation on some entity. The process may involve shaping, moulding or Loading and unloading. All these actions need power to transfer the entity from one state or condition into more valuable state or condition.

The second type of actions-transfer and positioning. In these cases, the product must generally be moved from one location to another during the series of processing steps.

2. Program of instructions: The actions performed by an automated process are defined by a set of instructions known as process. The programmed instructions determine the set of actions that is to be done automatically by the system. The program specifies what automated system should do and how its various components must function in order to accomplish the desired results.

3. Control system: The control element of the automated system executes the program of instructions.

The controls in an automated system can be:

(a) Closed loop

(b) Open loop.

(b) Closed loop control system: It is also known as a feedback control system. In this system the output variable is compared with an input parameter and any difference between the two is used to drive the output into agreement with input.

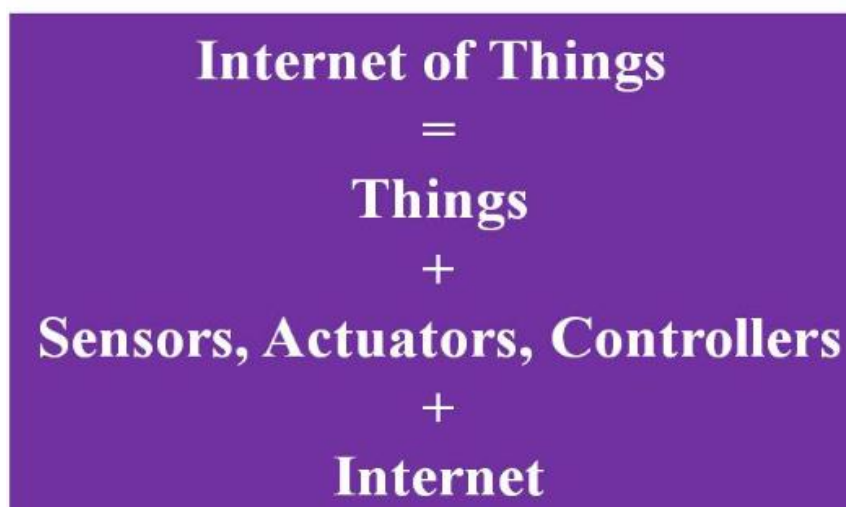
The Internet of Things (IOT)

The Internet of Things (IoT) describes the network of physical objects—“things”—that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet.

With more than 10 billion connected IoT devices today, experts are expecting this number to grow to 22 billion by 2025

The things receive inputs (using sensors) and transform them into data that is transmitted through Internet.

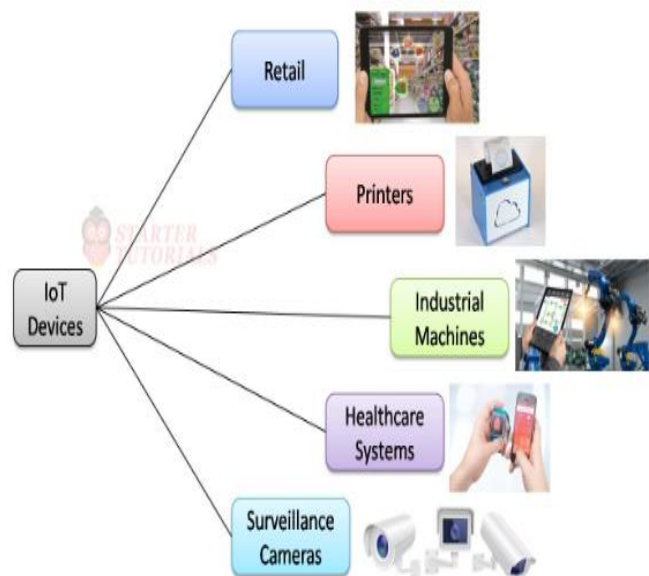
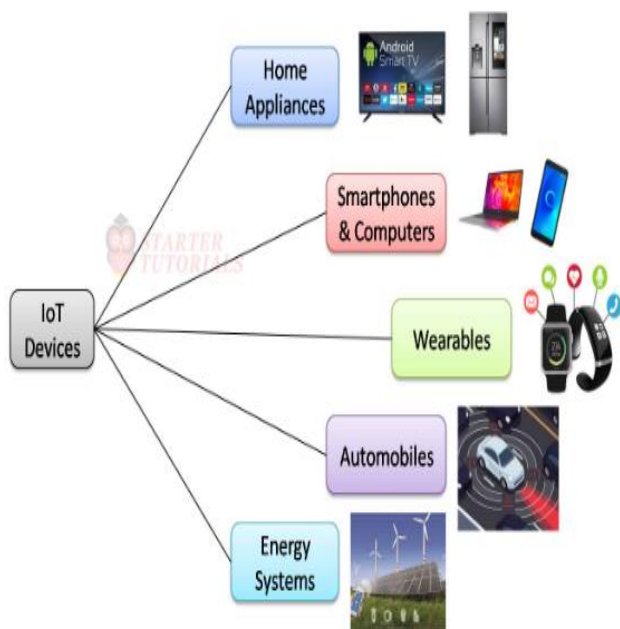
The things also can produce output or take action using actuators.



Some of the examples of IoT devices are:

- Home appliances: smart TV, smart refrigerator, smart AC, etc.

- Smart phones and computers
- Wearable's: smart watch, smart sensors, etc.
- Automobiles like self-driving cars
- Energy systems
- Retail : smart payment
- Printers
- Industrial machines
- Healthcare: smart watch, smart healthcare, etc.
- Surveillance: smart cameras, smart trackers, etc.



Advantages and Disadvantages of Internet of Things

Advantages	Disadvantages
It minimizes the human efforts	Can lead to network attacks as multiple devices are interconnected
It saves the time	As personal data are shared so it has privacy concerns
It enhances the data collection	As most of the work is performed by machine so it can lead to unemployment
The security is improved with the help of IoT	The complexity of the system is high
It is very efficient in resource utilization	There are high chances of the entire system to get corrupted easily
It reduces the usage of multiple electronic equipment.	There is a lack of international standardizations
It uses in traffic system very efficiently	As most of the work is performed using the data in cloud so it has high dependency on internet
We can also use IoT in healthcare Industry	It reduces mental and physical activity
Usage of sensors and other smart electronics devices save electricity consumption.	Lack of Security
	High dependency on internet so risk of malware attack

Advantages and Disadvantages of Internet of Things (Optional)

Advantages of IoT

Internet of things facilitates several advantages in our daily lives. Some of its advantages are given below:

- 1. Minimize human effort:** As IoT devices interact and communicate with each other, they can automate the tasks helping to improve the quality of a business's services and reducing the need for human intervention.

2. **Save time:** By reducing the human effort, it saves a lot of our time. Saving time is one of the primary advantages of using the IoT platform.
3. **Enhanced data collection:** Information is easily accessible, even if we are far away from our actual location, and it is updated frequently in real-time. Hence these devices can access information from anywhere at any time on any device.
4. **Improved security:** If we have an interconnected system, it can assist in the smarter control of homes and cities through mobile phones. It enhances security and offers personal protection.
5. **Efficient resource utilization:** We can increase resource utilization and monitor natural resources by knowing the functionality and how each device works.
6. **Useful for safety concerns:** It is helpful for safety because it senses any potential danger and warns users
7. **Useful in the healthcare industry:** Patient care can be performed more effectively in real-time without needing a doctor's visit. It gives them the ability to make choices as well as provide evidence-based care.

Disadvantages of IoT

As the Internet of things facilitates advantages, it also creates a significant set of drawbacks. Some of the IoT disadvantages are given below:

1. **Security issues:** IoT systems are interconnected and communicate over networks. So, the system offers little control despite any security measures, and it can lead to various kinds of network attacks.
2. **Privacy concern:** The IoT system provides critical personal data in full detail without the user's active participation.
3. **Increased unemployment:** Unskilled workers or even the skilled ones are at a high risk of losing their jobs, leading to high unemployment rates.
4. **The complexity of the system:** The designing, developing, maintaining, and enabling the extensive technology to IoT system is quite complicated.
5. **High chances of the entire system getting corrupted:** If there is a bug in the system, it is possible that every connected device will become corrupted.
6. **Lack of international standardizations:** As there is no international standard of compatibility for IoT, it is problematic for devices from different manufacturers to communicate with each other.
7. **High dependency on the internet:** They rely heavily on the internet and cannot function effectively without it.
8. **Reduced mental and physical activity:** Overuse of the internet and technology makes people ignorant because they rely on smart devices instead of doing physical work, causing them to become lethargic and inactive.

Characteristics of the Internet of Things(IoT)

IoT has various characteristics but most common are explained under,

1. Identity

Each IoT device has its unique identity. If it needs to access the data from specific device then its identification element is very helpful. The collaboration of name and number gives an identity to an internet device. Giving an identity to the device is an essential aspect of IoT. Identity helps to differentiate between various internet devices and select the device we want to send the command.

For Example, An **IP address** of a device is a **unique identifier** that represents a device in a network.

2. Dynamic and Self-Adapting

IOT devices should dynamically adapt themselves according to situations Assume a camera meant for the surveillance. It should be adaptable to work in different conditions and different light situations (morning, afternoon, night).

3. Self-Configuring – This is one of the most important characteristics of IoT.IoT devices are able to upgrade their software in accordance with requirements with a minimum of user participation. Additionally, they can set up the network, allowing for the addition of new devices to an already-existing network.

4. Connectivity

As the name suggests, it's all about the connectivity of 2 or more devices. This characteristics of IoT help to communicate with and share information between two or more devices. The IoT has created a world where everything is connected to communicate well and share user data. This connectivity also enables objects to be controlled remotely

5. Scalability –

The number of elements connected to the IoT zone is increasing day by day. Hence, an IoT setup should be capable of handling the massive expansion. The data generated as an outcome is enormous, and it should be handled appropriately

6. Safety

The basic idea of IoT is to connect everything to the internet and make the system easier for the users. However, when things are connected to the internet in such a way there is always a danger of the sensitive personal details of the users getting compromised. So safety is undoubtedly a crucial characteristic of IoT.

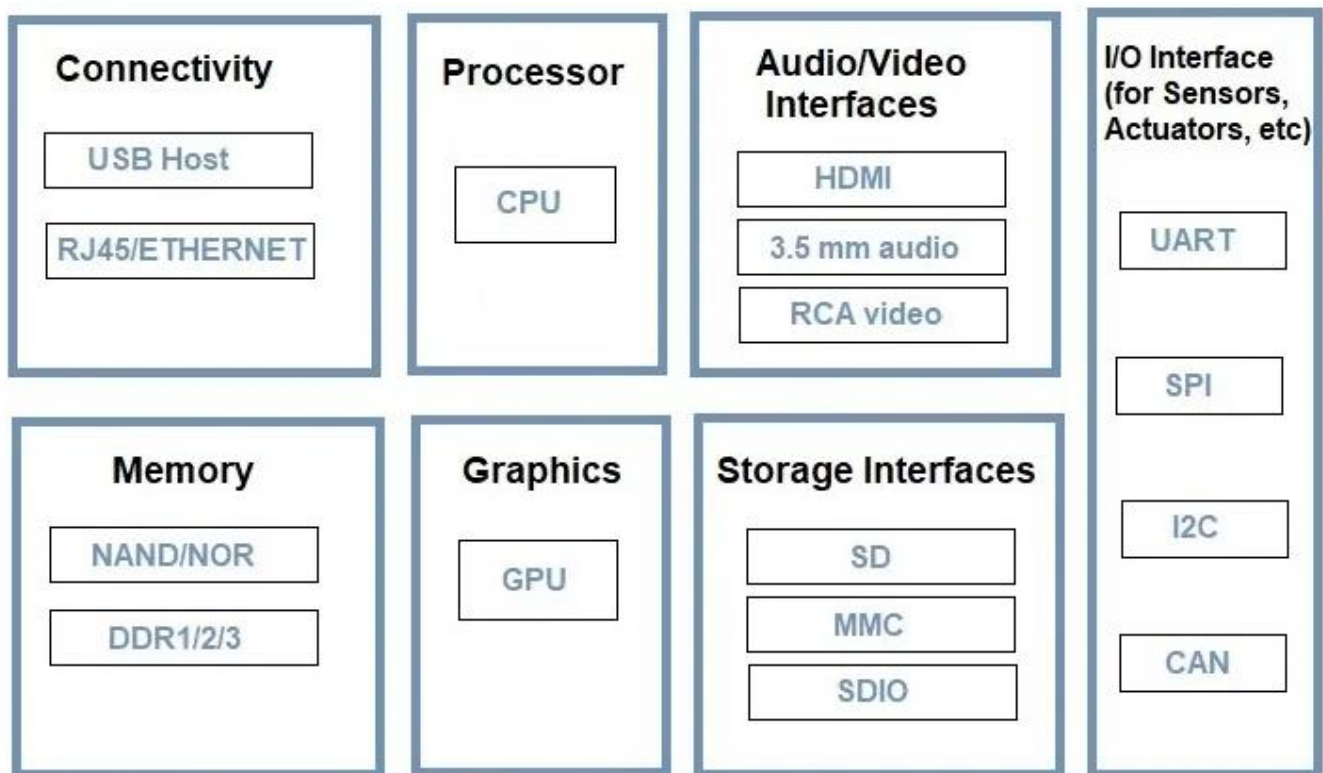
7. Architecture/Heterogeneity

The architecture of IoT is designed in a way that it is capable of supporting various devices, technologies, and protocols. Its main work is to confirm whether each connected device does not interfere with the other. This way, the safety and security of each device's data are maintained.

Physical Design of Internet of Things (IoT)

The physical design of an IoT system is referred to as the Things/Devices and protocols that are used to build an IoT system. All these things/Devices are called Node Devices and every device has a unique identity that performs remote sensing, actuating and monitoring work and the protocols that are used to establish communication between the Node devices and servers over the internet.

Generic block diagram of an IoT Device



Generic Block Diagram of IoT Devices

Connectivity

Devices like USB hosts and ETHERNET are used for connectivity between the devices and the server.

Processor

A processor like a CPU and other units are used to process the data. These data are further used to improve the decision quality of an IoT system.

Audio/Video Interfaces

An interface like HDMI and RCA devices is used to record audio and videos in a system. It shows all the information that the processor sends to the screen

Input/output interface

To give input and output signals to sensors, and actuators we use things like UART, SPI, CAN, etc.

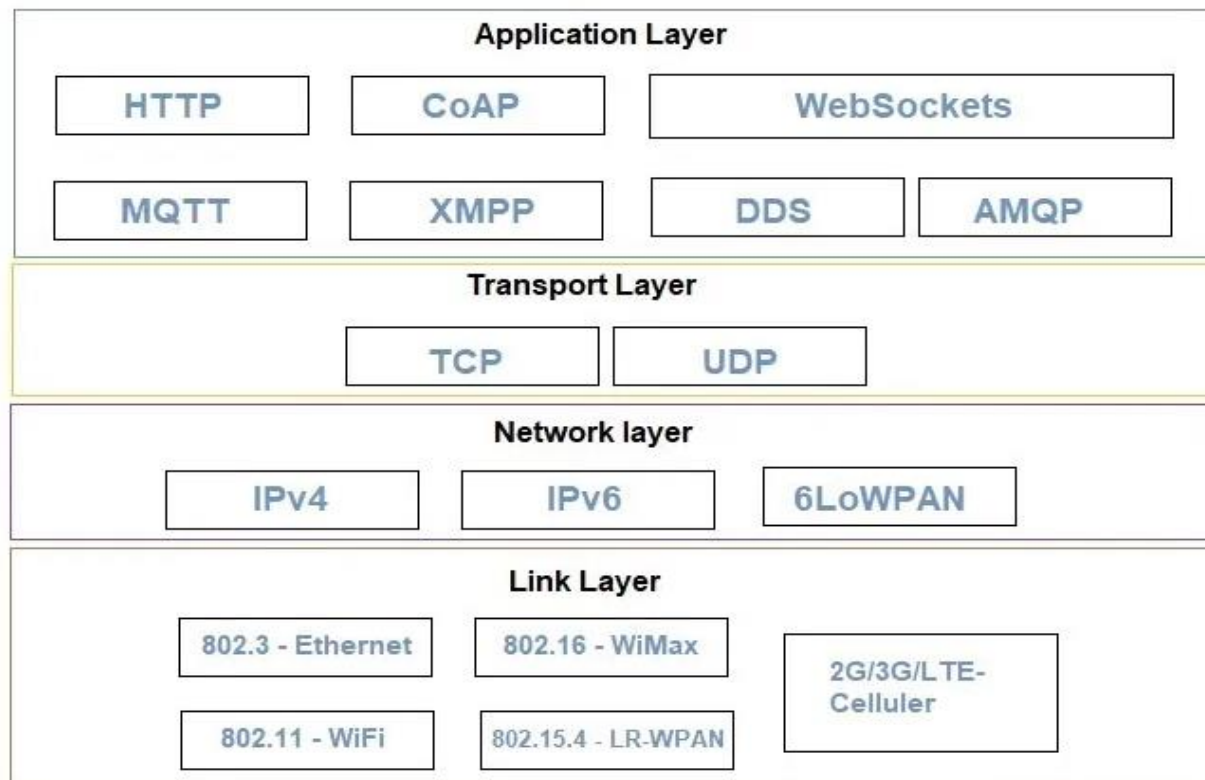
Storage Interfaces

Things like SD, MMC, and SDIO are used to store the data generated from an IoT device.

IoT Protocols

The set of rules governing all direct or indirect exchange of data between computers on a network. These rules are formulated at the application level and are used collectively to define how devices communicate irrespective of differences in their internal designs and operations.

IoT protocols help send commands and data between a network of devices controlled by sensors or other physical attributes like motion, temperature, or vibration. Network protocols help achieve reliable data transfer across all layers like the application, transport, network, and link layer.



Application Layer Protocols

On this layer, protocols use an application interface to define how the data can be sent over the network. These protocols include HTTP, XMPP, Web Socket, DDS, MQTT, and AMQP.

Transport Layer Protocols

This layer is responsible for data flow control and error handling, ensuring that there are rules in place to deal with errors. This layer also provides end-to-end message transfer capability, independent of the underlying network infrastructure. It provides essential connectivity between the two nodes on either end of the point-to-send-point-receive model used by key protocols such as TCP/IP.

Network Layer

This layer is used to send data from a source network to a destination network. For this, IPv4 and IPv6 protocols are used for host identification, which transfers data in packets.

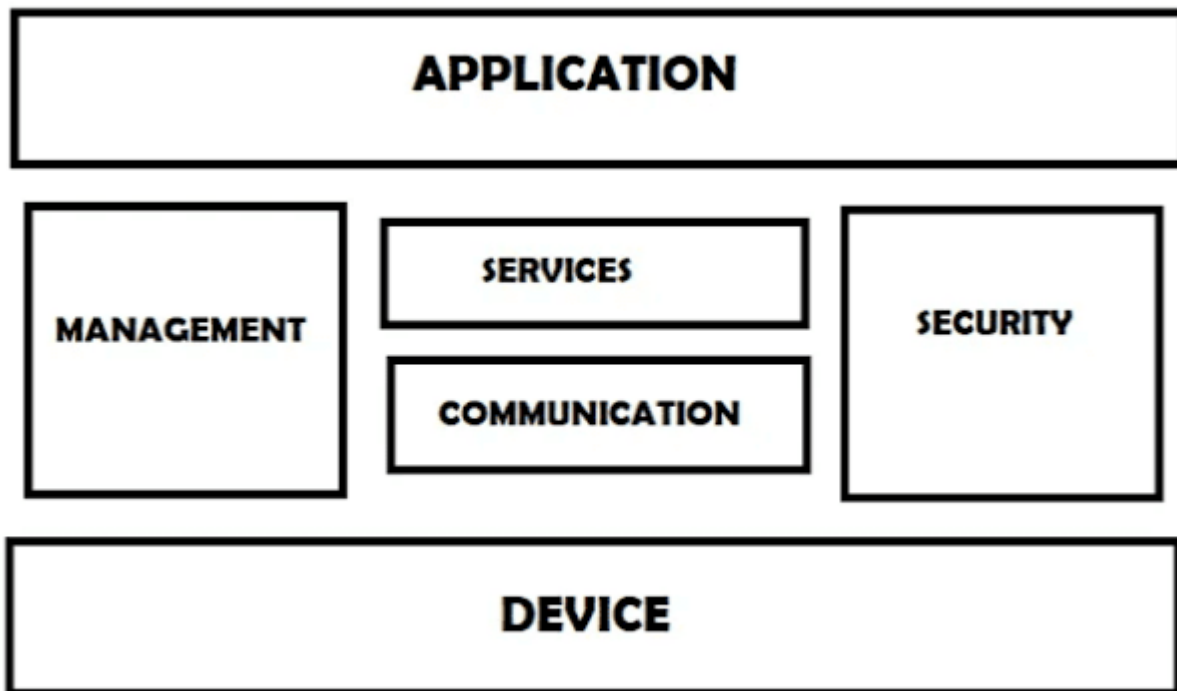
Link Layer

Link-layer protocols are the type of data transmission protocol used to help send data over the physical layer. They also determine how devices signal and code packets on the network.

Logical Design of Internet of Things (IoT)

IoT Functional blocks

An IoT system comprises of a number of functional blocks that provide the system the capabilities for identification, sensing, actuation, communication and management functional blocks are:



Application

It is an interface that provides a control system that use by users to view the status and analyze of system.

Management

This functional block provides various functions that are used to manage an IoT system.

Services

This functional block provides some services like monitoring and controlling a device and publishing and deleting the data and restoring the system.

Communication

This block handles the communication between the client and the cloud-based server and sends/receives the data using protocols.

Security

This block is used to secure an IoT system using some functions like authorization, data security, authentication, 2-step verification, etc.

Device

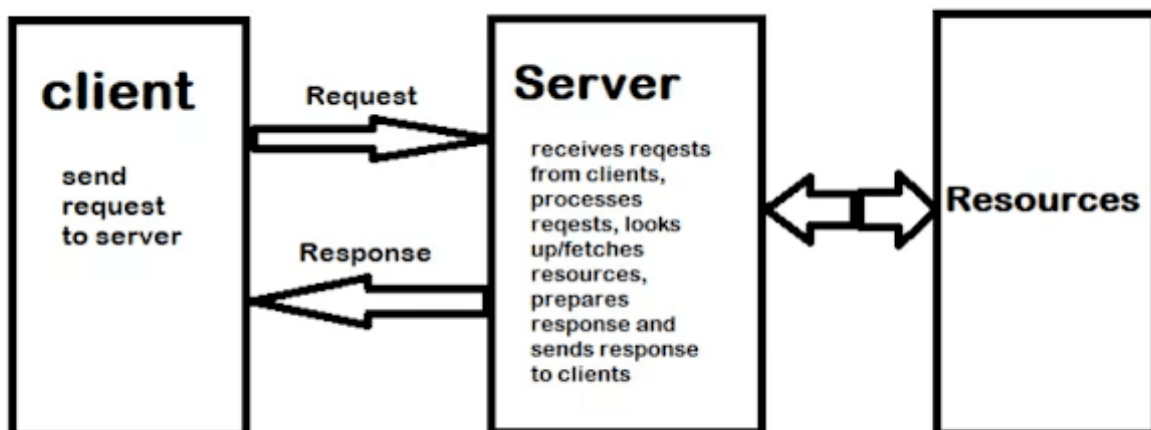
These devices are used to provide sensing and monitoring control functions that collect data from the outer environment.

IoT Communication Models

There are several different types of models available in an IoT system that is used to communicate between the system and server like the request-response model, publish-subscribe model, push-pull model, exclusive pair model, etc.

Request-Response Communication Model

This model is a communication model in which a client sends the request for data to the server and the server responds according to the request. When a server receives a request it fetches the data, retrieves the resources and prepares the response, and then sends the data back to the client.



Request-Response Communication Model

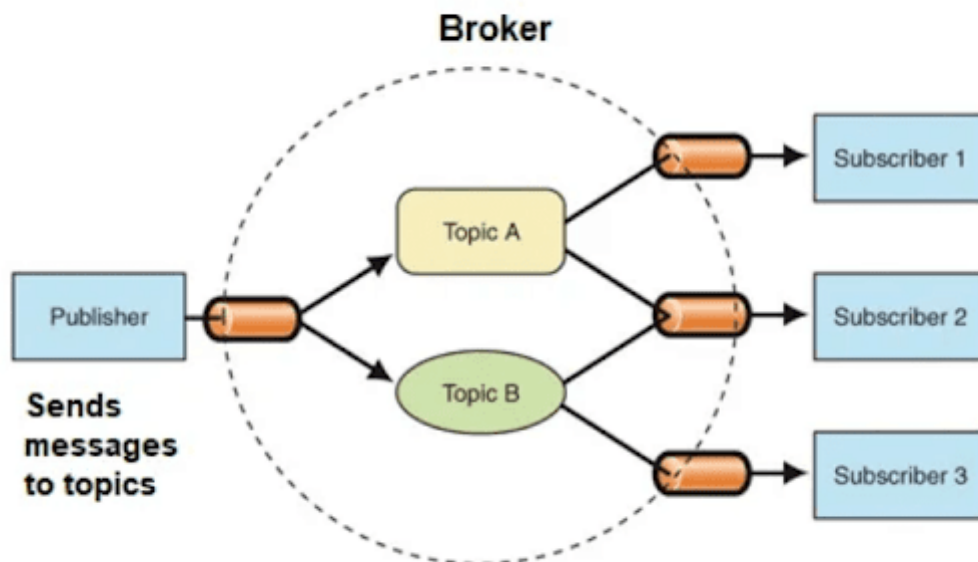
In simple terms, we can say that in the request-response model server send the response of equivalent to the request of the client. In this model, HTTP works as a request-response protocol between a client and server.

Example

When we search a query on a browser then the browser submits an HTTP request to the server and then the server returns a response to the browser (client).

Publish-Subscribe Communication Model

In this communication model, we have a broker between publisher and consumer. here publishers are the source of data but they are not aware of consumers. they send the data managed by the brokers and when a consumer subscribes to a topic that is managed by the broker and when the broker receives data from the publisher it sends the data to all the subscribed consumers.

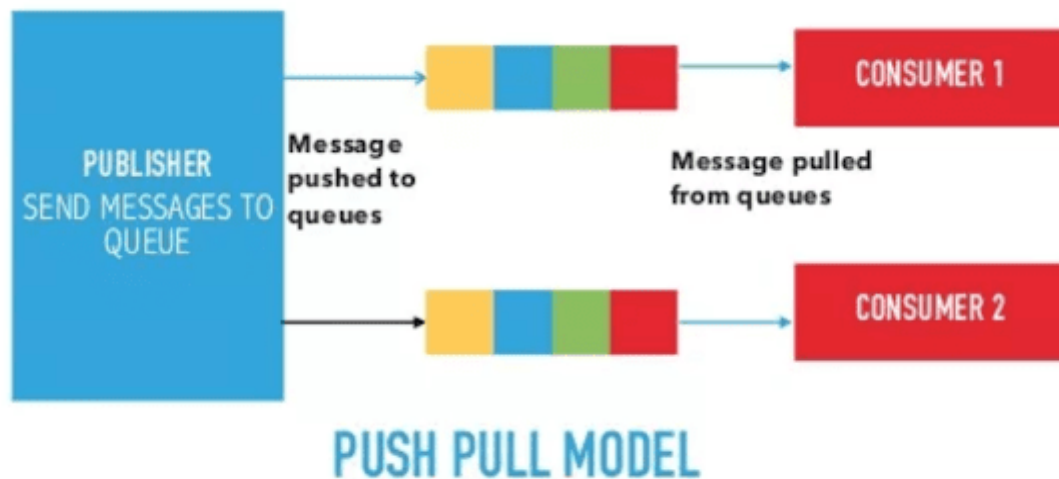


Example

On the website many times we subscribed to their newsletters using our email address. these email addresses are managed by some third-party services and when a new article is published on the website it is directly sent to the broker and then the broker sends these new data or posts to all the subscribers.

Push-Pull Communication Model

It is a communication model in which the data push by the producers in a queue and the consumers pull the data from the queues. here also producers are not aware of the consumers.



Example

When we visit a website we saw a number of posts that are published in a queue and according to our requirements, we click on a post and start reading it.

Exclusive Pair Communication Model

It is a bidirectional fully duplex communication model that uses a persistent connection between the client and server. Here first set up a connection between the client and the server and remain open until the client sends a close connection request to the server.



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